

Deep Hedging with Transaction Cost Model

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Abstract

In this study, we explored the impact of transaction costs on hedging strategies. The focus was on discussing the effects of transaction costs on changes in trading frequency and stock holding variations. We focused on three scenarios: no transaction fees, transaction fees in the Taiwanese market, and transaction fees in the U.S. market. Through Monte Carlo simulations, we compared the performance differences between Delta hedging and deep hedging under various transaction cost conditions. Finally, we used actual historical data to check whether the obtained results are same with the model predictions. The ultimate research findings revealed that the transaction costs not only significantly reduced the overall profitability of trading strategies but also noticeably changed the trading volume and strategy adjustment patterns in high-cost environments.

1 Introduction

Explores this project the hedging strategies using machine learning methods. HANS BUEHLER, LUKAS GONON, JOSEF TEICHMANN, AND BEN WOOD have provided rigorous mathematical proofs to demonstrate that the solution of this method converges to their desired results and have compared it with traditional hedging strategies, confirming its superior performance.[4]

It is well known that the stock market is not complete.[6] Therefore, many hedging strategies need to be improved such that they can work effectively in the real stock market. Transaction costs are an important consideration factor in the real market. With the presence of transaction costs, our hedging strategy may need to be modified. Therefore, I would like to observe how our hedging strategy changes when taking transaction costs into account.

I plan to use the deep hedging approach as my hedging strategy. In my previous work[2], I implemented the algorithm and compared its hedging effectiveness with delta hedging. However, I did not consider transaction costs at that time, and the hedging asset was generated using the Black-shole(BS) model. In this report, I aim to achieve the following:

1. I want to incorporate transaction costs and observe the changes in the hedging strategy. The transaction cost functions I want to discuss are:
 - 1.1. Taiwan market transaction costs: For each buy or sell transaction, a cost of stock price $\times$ number of shares $\times 0.1425\%$ $\times$ broker discount (since broker discount information is not publicly available, this factor will be omitted). [5]
 - 1.2. U.S. market transaction costs: For each buy or sell transaction, a cost of 0.01 to 0.02 of the total purchase price is charged.[3]
2. Use real historical data [1]into my hedging strategy and discuss whether this hedging strategy's effectiveness is significantly improved.

I am considering these transaction cost functions because they represent the actual amounts we are charged when trading in real life, making them the most practical and realistic choice.

2 Research methods

I will first study the impact of transaction costs on trading strategies. I assume it is under the BS-model, and I assume that I sold a call option on stock A with a strike price of 100 and an expiration

time of one month. The initial price of stock A is 100, $\mu = 0$, $\sigma = 0.2$. I generate 5000 paths and finally calculate the average return of this trading strategy.

The three trading strategies I want to compare are delta hedging, deep hedging with CVaR ($\lambda = 0.95$) trained 1000 times, and deep hedging with Entropic ($\lambda = 0.95$, the same as CVaR's λ) trained 10000 times (because its convergence rate is slower). Under different transaction cost ratios of 0, 0.001425, and 0.01, representing no fees, Taiwan's fees, and US fees respectively, what are the changes in stock holdings and the profit and loss (PnL) at the expiration date?

With the basic concepts of the BS-Model, I will discuss the real historical price data for comparison. I have selected the historical prices of Apple(APPL) and Amazan(AMZN) from 2012\01\01 to 2023\01\01, a total of 2739 datas. To have enough training data, I cut the data from 1 – 30 as one group, 2 – 31 as one group, 3 – 32 as one group, and so on. I use the first 2500 groups as training data and the remaining as test data. For convenience of calculation, I shift each data to start at a price of 100. The σ calculation should ideally use implied volatility, but since I can't obtain historical option prices, so I can't do this calculation. So I use the σ calculated from the previous 100 days of historical data for each data as the market σ for the next period. Under this market model, how do the three hedging strategies compare in terms of changes in stock holdings and the profit and loss (PnL) at expiration date on the historical data? When considering different transaction costs, what are the changes in stock holdings and the profit and loss (PnL) at expiration date for each strategy?

3 Results

1. Under the BS-model

1.1. Delta hedging with different cost

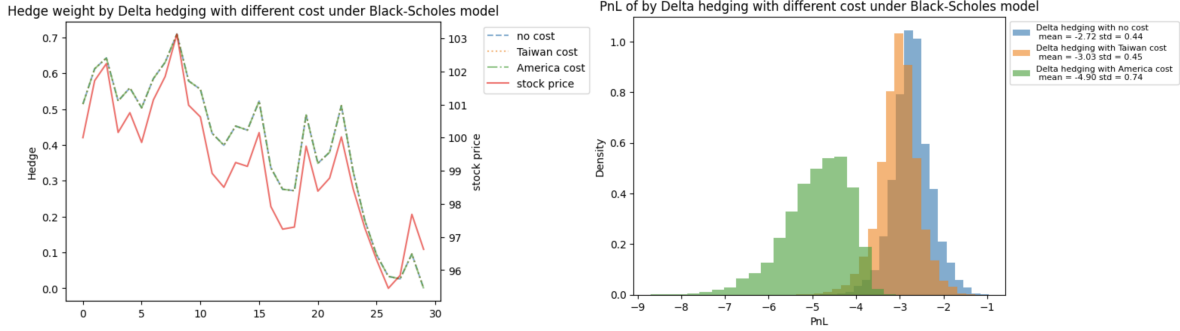


Figure 1: Asset's weight from Delta Hedging under the BS Model

Figure 2: PnL from Delta Hedging under the BS Model

For figure 1 and 2. Under delta hedging with transaction costs, the higher the fee ratio, the lower the PnL. The weights do not change at all. This result makes sense because the delta hedging strategy depends only on the delta of the call option and is independent of transaction costs. Therefore, the weights do not change. Given the same returns, higher transaction costs naturally lead to lower PnL.

1.2. Deep hedging(CVaR) with different cost

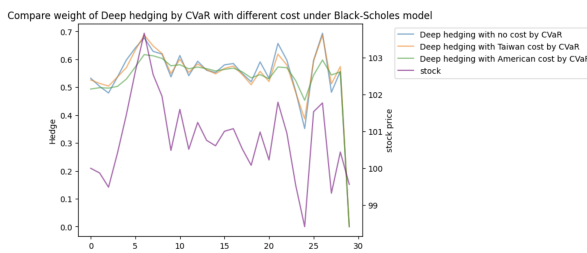


Figure 3: Asset's weight from Deep hedging(CVaR) under the BS Model

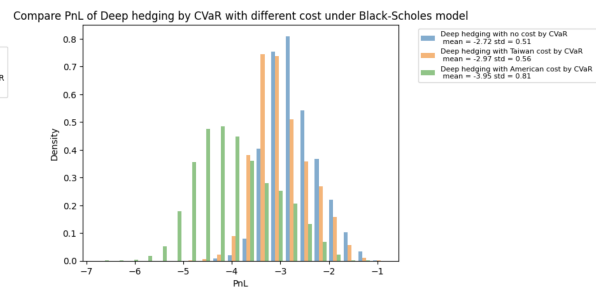


Figure 4: PnL from Deep hedging(CVaR) under the BS Model

For figure 3 and 4. Under deep hedging (CVaR) with transaction costs, the higher the fee ratio, the lower the PnL, the fluctuation in weights also becomes smaller and the average PnL decreases as well. This result makes sense because when transaction costs are higher, the trading volume is reduced to lower the transaction costs in order to maintain PnL.

1.3. Deep hedging(Entropic) with different cost

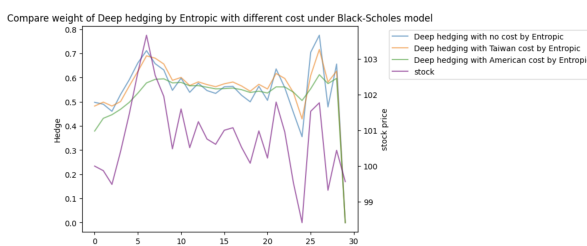


Figure 5: Asset's weight from Deep hedging(Entropic) under the BS Model

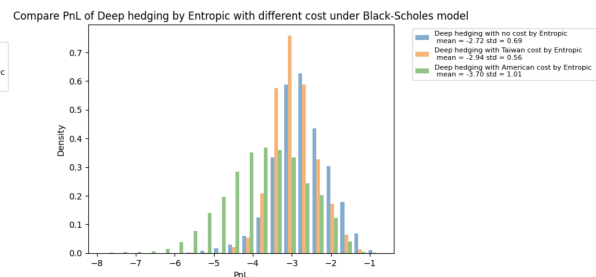


Figure 6: PnL from Deep hedging(Entropic) under the BS Model

For figure 5 and 6. Under deep hedging (Entropic) with transaction costs, the higher the fee ratio, the lower the PnL, the fluctuation in weights also becomes smaller and the average PnL decreases as well. This result makes sense because when transaction costs are higher, the trading volume is reduced to lower the transaction costs in order to maintain PnL.

1.4. No cost with different strategies



Figure 7: Asset's weight from no cost under the BS Model

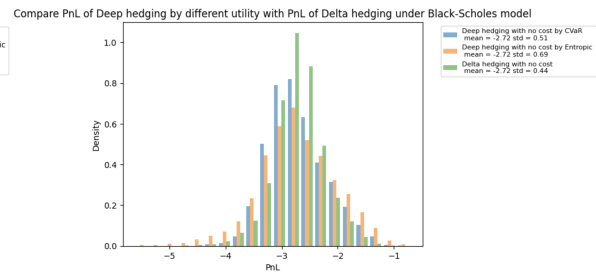


Figure 8: PnL from no cost under the BS Model

For figure 7 and 8. In the case without transaction costs, although the mean PnL of delta hedging is the same as the other two strategies, its standard deviation is smaller. Therefore, delta hedging performs better than the other two strategies. As for the weights, they can be said to be largely similar without significant differences.

1.5. Taiwan cost with different strategies

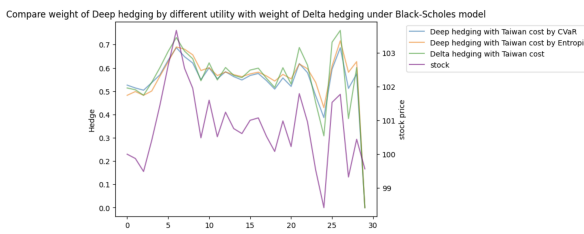


Figure 9: Asset's weight from Taiwan cost under the BS Model

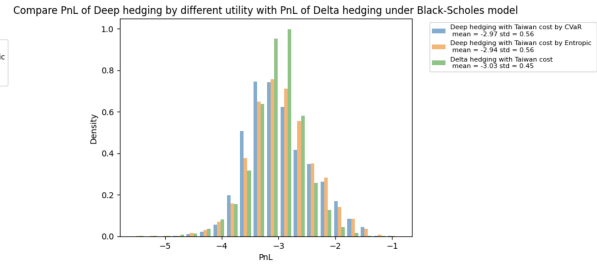


Figure 10: PnL from Taiwan cost under the BS Model

For figure 9 and 10. In the case with Taiwan's transaction costs, although the mean PnL of delta hedging is lower than the other two strategies, its standard deviation is still smaller. Therefore, it is unclear which strategy performs better. As for the weights, it is observed that deep hedging trades even less than delta hedging.

1.6. American cost with different strategies

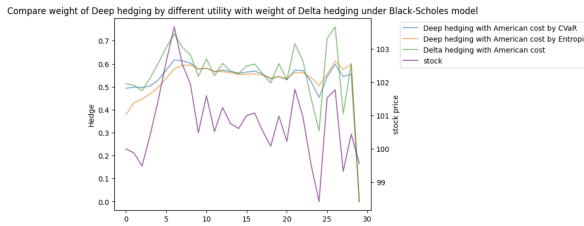


Figure 11: Asset's weight from American cost under the BS Model

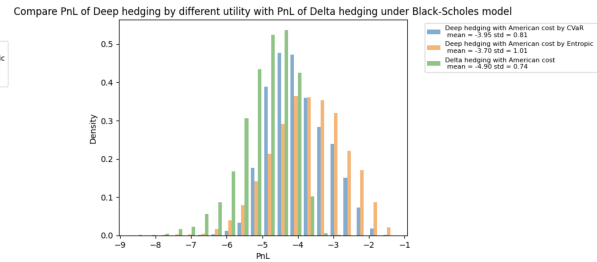


Figure 12: PnL from American cost under the BS Model

For figure 11 and 12. In the case with American transaction costs, since the mean PnL of delta hedging is much lower than the other two strategies, deep hedging performs better. As for the weights, it is clearly observed that deep hedging trades significantly less than delta hedging.

2. Under the APPL history data

2.1. Delta hedging with different cost

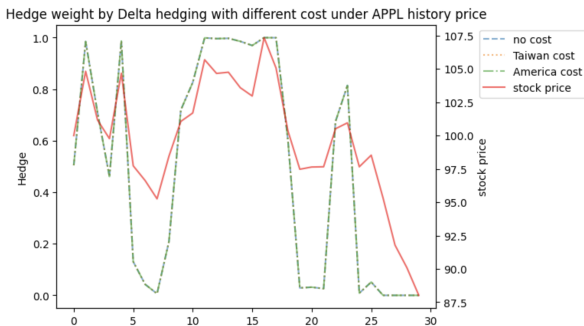


Figure 13: Asset's weight from Delta Hedging under the APPL history data

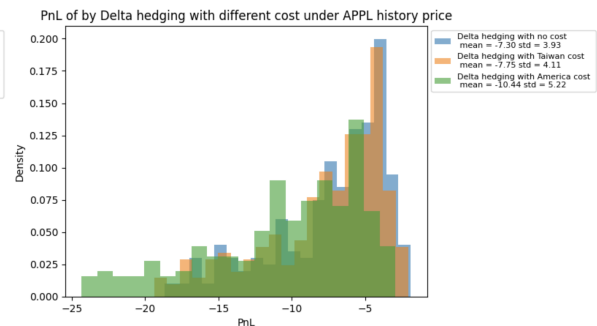


Figure 14: PnL from Delta Hedging under the APPL history data

For figure 13 and 14. Under delta hedging with transaction costs, the conclusion is the same as 1.1.

2.2. Deep hedging(CVaR) with different cost

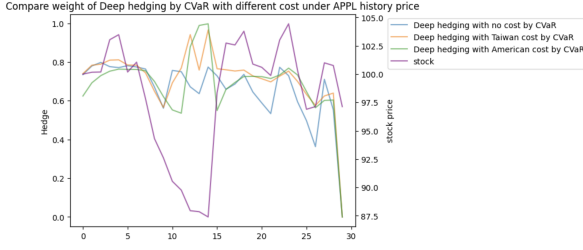


Figure 15: Asset's weight from Deep hedging(CVaR) under the APPL history data

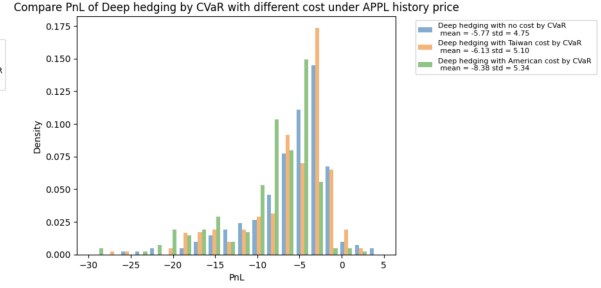


Figure 16: PnL from Deep hedging(CVaR) under the APPL history data

For figure 15 and 16. Under deep hedging (CVaR) with transaction costs, the higher the fee ratio, the lower the PnL, and the standard deviation is also larger. However, the fluctuation in weights does not decrease due to higher transaction costs.

2.3. Deep hedging(Entropic) with different cost

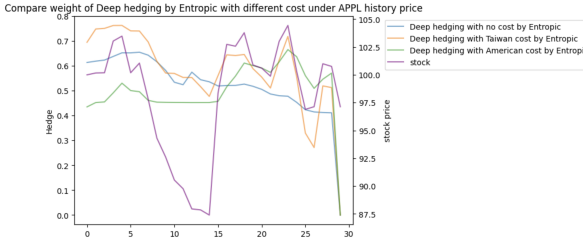


Figure 17: Asset's weight from Deep hedging(Entropic) under the APPL history data

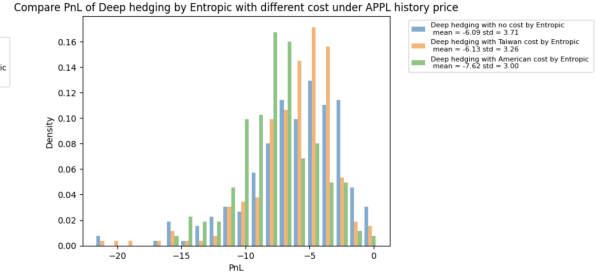


Figure 18: PnL from Deep hedging(Entropic) under the APPL history data

For figure 17 and 18. Under deep hedging (Entropic) with transaction costs, the higher the fee ratio, the lower the PnL, but the standard deviation is actually smaller. Regarding the weights, similar to the discussion in 2.2, the fluctuations do not decrease due to higher transaction costs.

2.4. No cost with different strategies

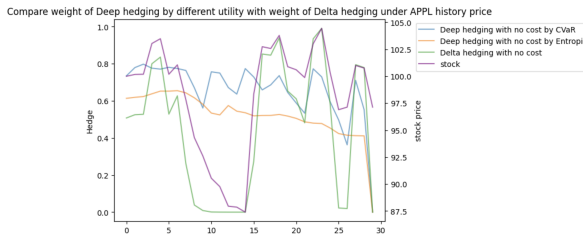


Figure 19: Asset's weight from no cost under the APPL history data

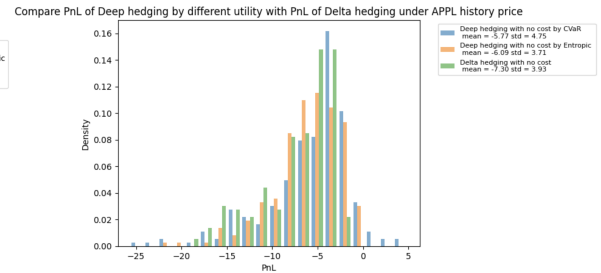


Figure 20: PnL from no cost under the APPL history data

For figure 19 and 20. In the case without transaction costs, since the mean PnL of delta hedging is much lower than the other two strategies, deep hedging performs better. As for the weights, it is clearly observed that deep hedging trades significantly less than delta hedging. Therefore, this result differs from the one in 1.4.

2.5. Taiwan cost with different strategies

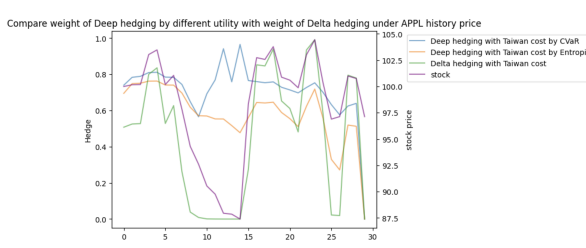


Figure 21: Asset's weight from Taiwan cost under the APPL history data

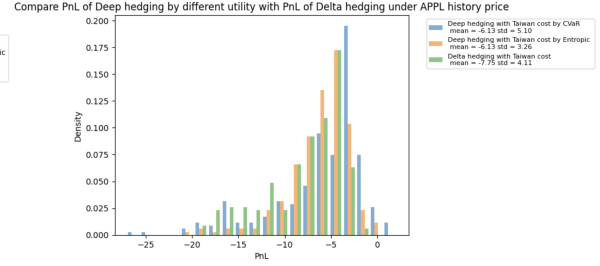


Figure 22: PnL from Taiwan cost under the APPL history data

For figure 21 and 22. In the case with Taiwan's transaction costs, the conclusion is the same as 2.4 deep hedging performs better due to higher PnL. However, upon closer inspection of the weight changes, there are some peculiarities. The weights sometimes change in the opposite direction to the price movements. This aspect will be discussed in the conclusion as to why this is the case.

2.6. American cost with different strategies

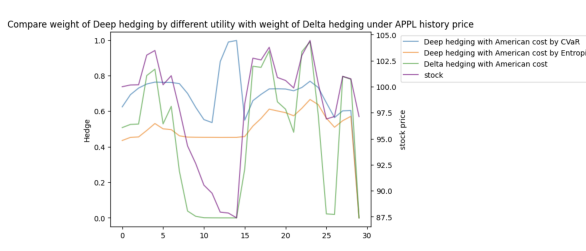


Figure 23: Asset's weight from American cost under the APPL history data

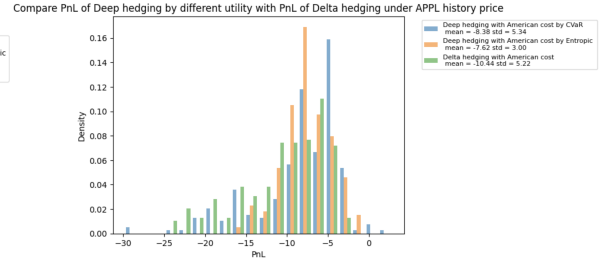


Figure 24: PnL from American cost under the APPL history data

For figure 23 and 24. In the case with American transaction costs, the conclusion is the same as 2.5.

3. Under the AMZN history data

3.1. Delta hedging with different cost



Figure 25: Asset's weight from Delta Hedging under the AMZN history data



Figure 26: PnL from Delta Hedging under the AMZN history data

For figure 25 and 26. Under delta hedging with transaction costs, the conclusion is the same as 1.1. However, upon closer inspection of the weight changes, there are some peculiarities.

The weights do not fully fluctuate in line with the price movements. This aspect will be discussed in the conclusion as to why this is the case.

3.2. Deep hedging(CVaR) with different cost



Figure 27: Asset's weight from Deep hedging(CVaR) under the AMZN history data

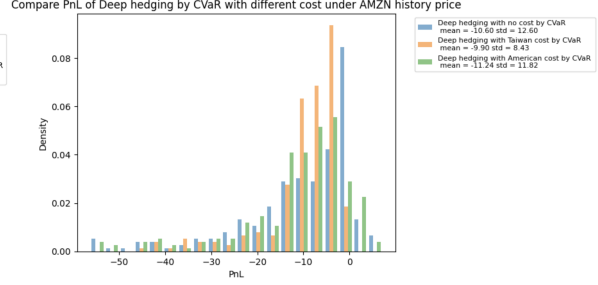


Figure 28: PnL from Deep hedging(CVaR) under the AMZN history data

For figure 27 and 28. Under deep hedging (CVaR) with transaction costs, the higher the fee ratio should lead to lower PnL. However, a peculiar phenomenon occurred where under Taiwan's transaction costs, the mean and standard deviation were actually better. Additionally, the weights sometimes changed in the opposite direction to the price movements.

3.3. Deep hedging(Entropic) with different cost



Figure 29: Asset's weight from Deep hedging(Entropic) under the AMZN history data

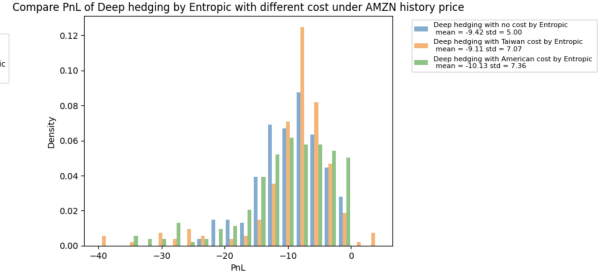


Figure 30: PnL from Deep hedging(Entropic) under the AMZN history data

For figure 29 and 30. Under deep hedging (Entropic) with transaction costs, the PnL results appear normal this time, with higher fee ratio leads to lower PnL. However, there is an abnormal phenomenon observed in the weights, where straight line patterns emerge. This is a problem that needs to be solved.

3.4. No cost with different strategies

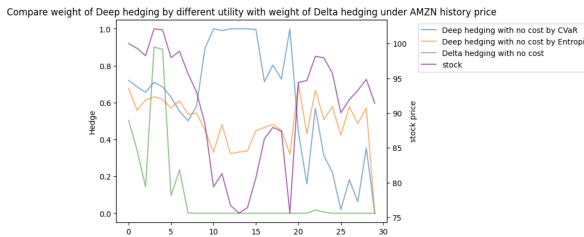


Figure 31: Asset's weight from no cost under the AMZN history data

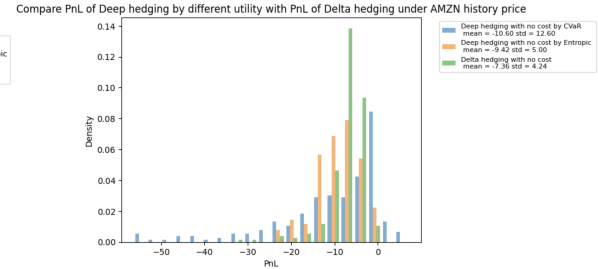


Figure 32: PnL from no cost under the AMZN history data

For figure 31 and 32. In the case without transaction costs, delta hedging has a larger mean PnL and smaller standard deviation, therefore the delta hedging strategy outperforms deep

hedging. They also have a problem. The weights of deep hedging do not fully fluctuate in line with the price movements.

3.5. Taiwan cost with different strategies

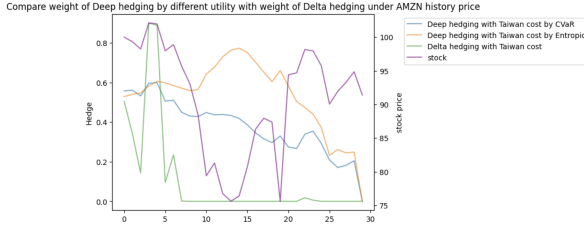


Figure 33: Asset's weight from Taiwan cost under the AMZN history data

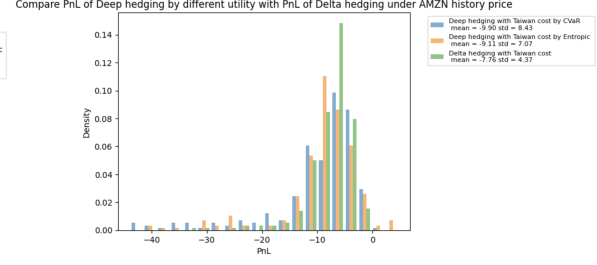


Figure 34: PnL from Taiwan cost under the AMZN history data

For figure 33 and 34. In the case with Taiwan's transaction costs, the conclusion is the same as 3.4.

3.6. American cost with different strategies

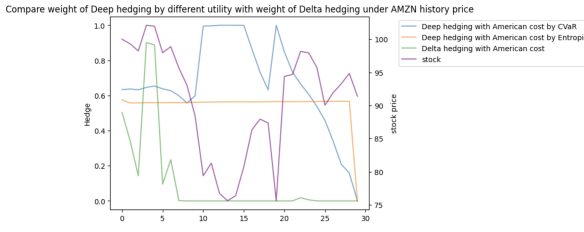


Figure 35: Asset's weight from American cost under the AMZN history data

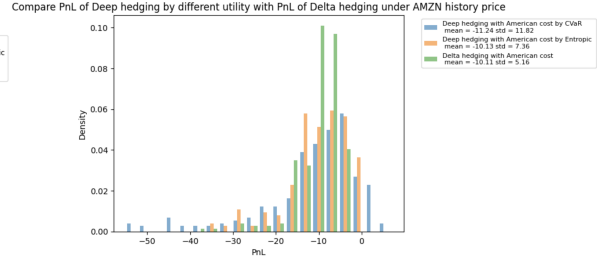


Figure 36: PnL from American cost under the AMZN history data

For figure 35 and 36. In the case with American transaction costs, the conclusion is the same as 3.4.

4 Concluding result

1. Under transaction costs, no matter which strategies we choose, the PnL decreases.
2. With higher transaction costs, the changes in trading volume for deep hedging become less frequent.
3. Under transaction costs, the performance of delta hedging becomes worse compared to the other two strategies.
4. When incorporating real historical data, and comparing to delta hedging, deep hedging should perform better when we consider the transaction costs. This is the expected result we get from we use the BS-model to test first, but it may not hold true in reality. I think the issue lies with σ , because the σ from the previous period does not necessarily equal the σ for the next period. So using the previous period's σ for prediction can lead to peculiarities, such as delta hedging sometimes outperforming deep hedging, weights not fully fluctuating with price movements under delta hedging, and weights sometimes moving opposite to price movements under deep hedging.
5. Using around 10 years of historical data for training, the performance for AMZN is noticeably worse. This may be related to AMZN's stock price declining in 2022-2023 while APPL's continued rising. The model training did not capture this signal, leading to many abnormal results for deep

hedging. To deal with this, more diverse data and increased training iterations may be needed to strengthen the model.

After considering transaction costs, this function model will be more aligned with the real market, but there are still many factors that need to be taken into account, such as liquidity issues, other trading restrictions, and market impact. I have not yet incorporated these factors into my program. Furthermore, the above conclusions still have some issues that need to be addressed. I hope that one day I can fill this gap.

References

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